

Victoria Line Train Service Optimisation

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1. Introduction

The aim of this project is to determine the optimal train allocation across services on the Victoria Line of the London Underground to minimise passenger overcrowding, using TfL NUMBAT data. An integer linear programming model is developed to find the minimum number of trains that can be dispatched in each interval and direction whilst maintaining zero overcrowding.

The initial model however only minimised overcrowding. A sensitivity analysis, investigating the effect of a substantial reduction in train capacity revealed a solution where overcrowding was zero could still be found. Therefore the model was revised to minimise train dispatches and a sensitivity analysis was performed, investigating the effect of a decrease in train capacity on the minimum number of trains that can be dispatched, whilst maintaining overcrowding at zero.

The model assumes a fixed number of cars per train and average capacity per car. Trains on the Victoria Line consist of 8 cars with an average capacity per car of 107, 856 is assumed to be the average capacity per train. The Victoria Line fleet size is assumed to be 47 trains and is incorporated as a fleet constraint in the model.

2. Data

2.1 Data source

The model uses data from TfL NUMBAT 2025

<https://crowding.data.tfl.gov.uk/>.

The data on a typical autumn monday (NBT25MON) was chosen and Link Loads data was used.

Information used in determining the model constraints was gathered from the following websites:

1. <https://tfl.gov.uk/corporate/transparency/freedom-of-information/foi-request-detail?referenceId=FOI-4157-2425>
2. <https://content.tfl.gov.uk/06-lu-rolling-stock.pdf>
3. <https://mycommute.london/victoria>

2.2 Data preparation

In the data preparation phase columns not relevant to the model were dropped and the Victoria Line was selected. Then the data frame was converted into a long format, time intervals were initially arranged in columns but the process converted them to rows. Finally the order link numbers were then adjusted - they were kept the same for NB but reversed for SB as instead of order the numbers were going to be allocated to links.

3. Mathematical Model

3.1 Assumptions

The NUMBAT data used 15 minute intervals, which significantly influenced the design of the model. The travel time between each station is roughly 2 minutes and the total time it takes to reach one end of the line to the other is approximately 31 minutes. However for simplicity this was assumed to be 30 minutes (and therefore 2 intervals).

The model modeled trains as completing a cycle around the line (starting from one end, reaching the end, turning around and then reaching the other end). However turnaround time was not taken into consideration in the model and it was instead assumed it would take 60 minutes (4 intervals) for a train to complete a full cycle.

To remain consistent with the 15 minute intervals, in each direction the line was split into halves from station. Four trains are assumed to be in circulation at $t = 0$, one in each half. This was to ensure overcrowding remained at zero at all times.

Additionally it is assumed all trains are identical with a fixed capacity of 856.

Link load was used to determine passenger demand and by setting overcrowding equal to zero, all passenger demand must be accommodated for in each interval.

Finally it is assumed there are no delays or cancellations.

3.2 Model 1

Let $t = 1, \dots, T$ denote each 15-minute interval

Let $I = 1, \dots, 15$ where $i \in I$ denotes the link number

1. Brixton - Stockwell
2. Stockwell - Vauxhall
3. Vauxhall - Pimlico
4. Pimlico - Victoria
5. Victoria - Green Park
6. Green Park - Oxford Circus
7. Oxford Circus - Warren Street
8. Warren Street - Euston
9. Euston - King's Cross St Pancras
10. King's Cross St Pancras - Highbury & Islington
11. Highbury & Islington - Finsbury Park
12. Finsbury Park - Seven Sisters
13. Seven Sisters - Tottenham Hale
14. Tottenham Hale - Blackhorse Road
15. Blackhorse Road - Walthamstow Central

Decision variables

$O_{i,t}^d$ – overcrowding in direction $d \in \{NB, SB\}$ at link i during interval t

x_t^{NB} – the number of trains dispatched from Brixton during interval t

x_t^{SB} – the number of trains dispatched from Walthamstow Central during interval t

Parameters

$L_{i,t}^d$ – the number of passengers travelling on link i in direction $d \in \{NB, SB\}$

c – capacity per train

F – fleet size

Objective function

$$\text{Minimise } \sum_{d \in \{\text{NB}, \text{SB}\}} \sum_{i=1}^{15} \sum_t^T O_{i,t}^d$$

Train circulation

Trains are modeled following a cyclical structure. Each train is either dispatched at Brixton or Walthamstow Central and follows the route until the end of the line and then turns around to reach the other end of the line, thus completing a cycle. It is assumed it takes 30 minutes to travel the full line, so takes 60 minutes to complete a cycle.

The line is split into halves to be compatible with the 15-minute intervals used in the NUMBAT dataset.

It is also assumed at $t = 0$ there is one train in circulation in each half of the line in each direction.

$$\begin{aligned} x_t^{NB} &= x_t^{SB} = 0 \text{ for } t \leq -2 \\ x_t^{NB} &= x_t^{SB} = 1 \text{ for } t \in \{-1, 0\} \end{aligned}$$

Capacity constraints

Link load is the demand on trains between platforms. The model defines overcrowding as the excess passenger demand over the provided capacity in the relevant link and interval.

$$O_{i,t}^{NB} \geq L_{i,t}^{NB} - c(x_t^{NB} + x_{t-2}^{SB}), \forall (i \leq 8 \cap t > 2)$$

$$O_{i,t}^{NB} \geq L_{i,t}^{NB} - c(x_{t-1}^{NB} + x_{t-3}^{SB}), \forall (i \geq 9 \cap t > 3)$$

$$O_{i,t}^{SB} \geq L_{i,t}^{SB} - c(x_t^{SB} + x_{t-2}^{NB}), \forall (i \geq 9 \cap t > 2)$$

$$O_{i,t}^{SB} \geq L_{i,t}^{SB} - c(x_{t-1}^{SB} + x_{t-3}^{NB}), \forall (i \leq 8 \cap t > 3)$$

Fleet constraint

$$\sum_{\tau=t-3}^t (x_{\tau}^{NB} + x_{\tau}^{SB}) \leq F, \forall t$$

Frequency constraint

Peak frequency (the most trains on the network - in one direction) is 36 trains per hour. Since the model used 15 minute intervals we have

$$36 \times \frac{15}{60} = 9$$

$$x_t^{NB} \leq 9 \text{ and } x_t^{SB} \leq 9$$

Integer constrain

$$x_t^d \in \mathbb{Z}_{\geq 0}$$

3.3 Model 2

The second model retains the train circulation, fleet and frequency constraints from model 1. However it removes the overcrowding constraint, essentially setting it equal to zero so it requires passenger demand to be no greater than available capacity.

Decision variables

x_t^{NB} – the number of trains dispatched from Brixton during interval t

x_t^{SB} – the number of trains dispatched from Walthamstow Central during interval t

Parameters

$L_{i,t}^d$ – the number of passengers travelling on link i in direction $d \in \{NB, SB\}$

c – capacity per train

F – fleet size

Objective function

$$\text{Minimise } \sum_{d \in \{NB, SB\}} \sum_{t=1}^T x_t^d$$

Constraints

$$x_t^{NB} = x_t^{SB} = 0 \text{ for } t \leq -2$$

$$x_t^{NB} = x_t^{SB} = 1 \text{ for } t \in \{-1, 0\}$$

$$L_{i,t}^{NB} \leq c(x_t^{NB} + x_{t-2}^{SB}), \forall (i \leq 8 \cap t > 2)$$

$$L_{i,t}^{NB} \leq c(x_{t-1}^{NB} + x_{t-3}^{SB}), \forall (i \geq 9 \cap t > 3)$$

$$L_{i,t}^{SB} \leq c(x_t^{SB} + x_{t-2}^{NB}), \forall (i \geq 9 \cap t > 2)$$

$$L_{i,t}^{SB} \leq c(x_{t-1}^{SB} + x_{t-3}^{NB}), \forall (i \leq 8 \cap t > 3)$$

$$4 + \sum_{\tau=t-3}^t (x_{\tau}^{NB} + x_{\tau}^{SB}) \leq F, \forall t$$

$$x_t^{NB} \leq 9$$

$$x_t^{SB} \leq 9$$

$$x_t^d \in \mathbb{Z}_{\geq 0}$$

4. Optimisation Approach

The model was implemented in Python using Pyomo and solved using GLPK for the first model. However the solution time became impractical for the second model so HiGHS was used instead. The initial model was implemented and a solution where overcrowding equaled zero was found for the baseline capacity of 856. However after adjusting the capacity down to even 428, the model still found a solution where overcrowding was equal to zero. Therefore the model was instead adjusted to minimise the trains dispatches needed for overcrowding to remain at zero.

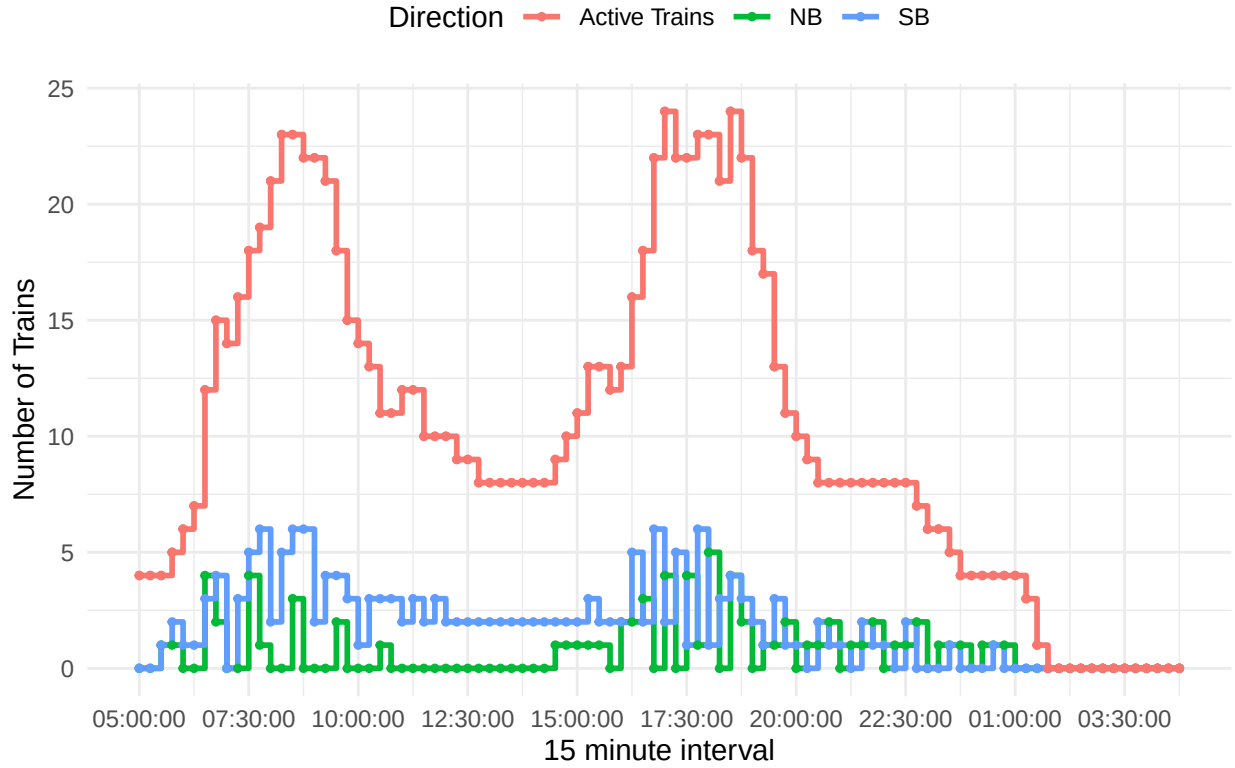
5. Results

5.1 Optimal dispatch schedule and Active fleet

Table 1: First 15 intervals of the optimised train dispatch schedule at capacity 856

Time	t	NB	SB	Trains_active
0500-0515	1	0	0	4
0515-0530	2	0	0	4
0530-0545	3	1	1	4
0545-0600	4	1	2	5
0600-0615	5	0	1	6
0615-0630	6	0	1	7
0630-0645	7	4	3	12
0645-0700	8	2	4	15
0700-0715	9	0	0	14
0715-0730	10	0	3	16
0730-0745	11	4	5	18
0745-0800	12	1	6	19
0800-0815	13	0	2	21
0815-0830	14	0	5	23
0830-0845	15	3	6	23

Optimised Train Dispatches

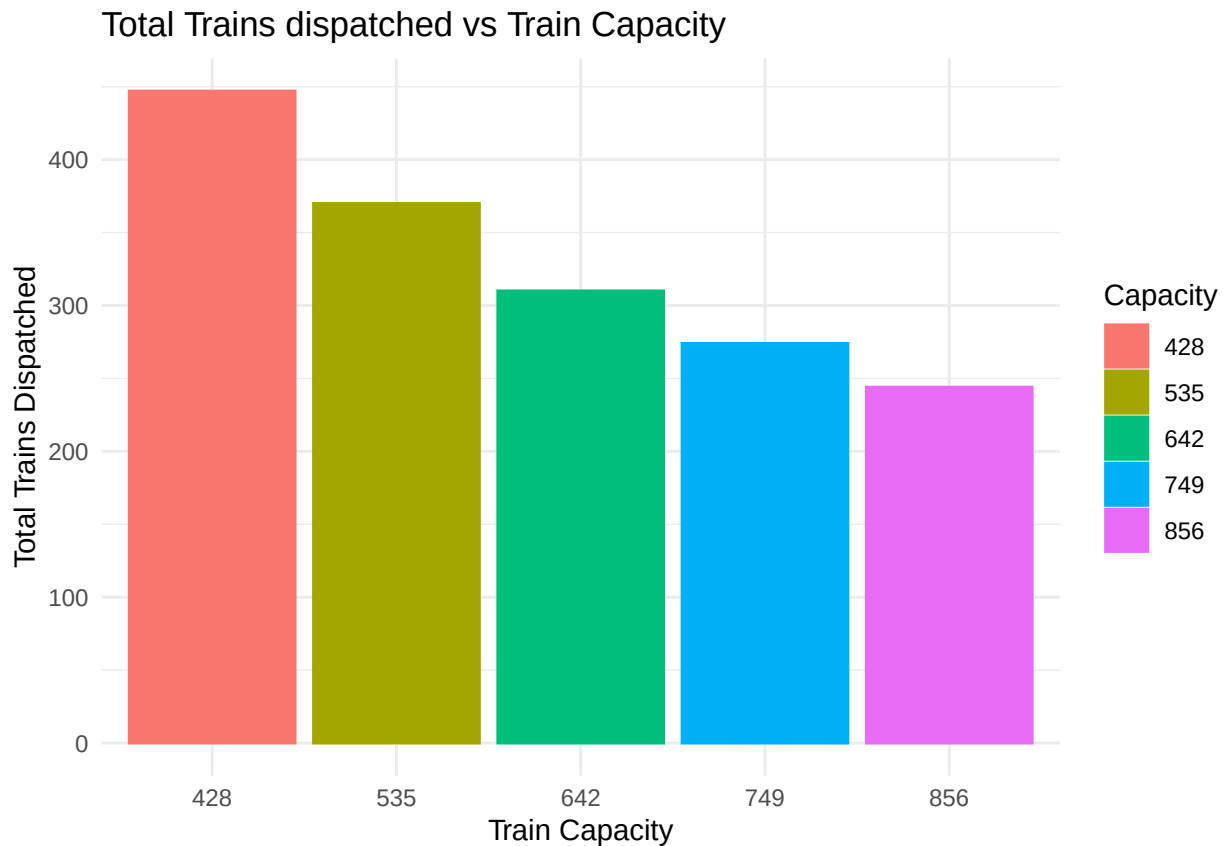


The plot shows the optimised number of trains in circulation in each 15 minute interval (assuming capacity is 856). The number of trains being dispatched fluctuates, although the number of trains in circulation

increases to meet peak demand in the morning. The number of active trains peaks around 08:00/08:30 at 23 in circulation. After the initial rush, link load falls so the number of trains being dispatched during each interval falls and so does the number of active trains decreasing. The number of active trains continues to fall until just after 12:30, where it remains at 8 trains active on the line until just before 15:00. From this point the number of trains dispatched in both directions increases and the number of trains in circulation increases and peaks at 24 at around 17:00, and begins to fall again at around 19:00. The number of active trains continues to fall until just after 20:00, when it then plateaus at 8 in circulation, then falling again after 22:30, eventually falling to 0 after 01:00 in the morning. It can be seen that more trains are dispatched on the Southbound line, especially during peak times. Additionally it is only after 20:00 that during some intervals no trains are dispatched on the Southbound line (with the exception of the first two intervals after 05:00). However on the Northbound line between the rough hours of 11:00 and 14:30 no trains are dispatched.

6. Sensitivity Analysis

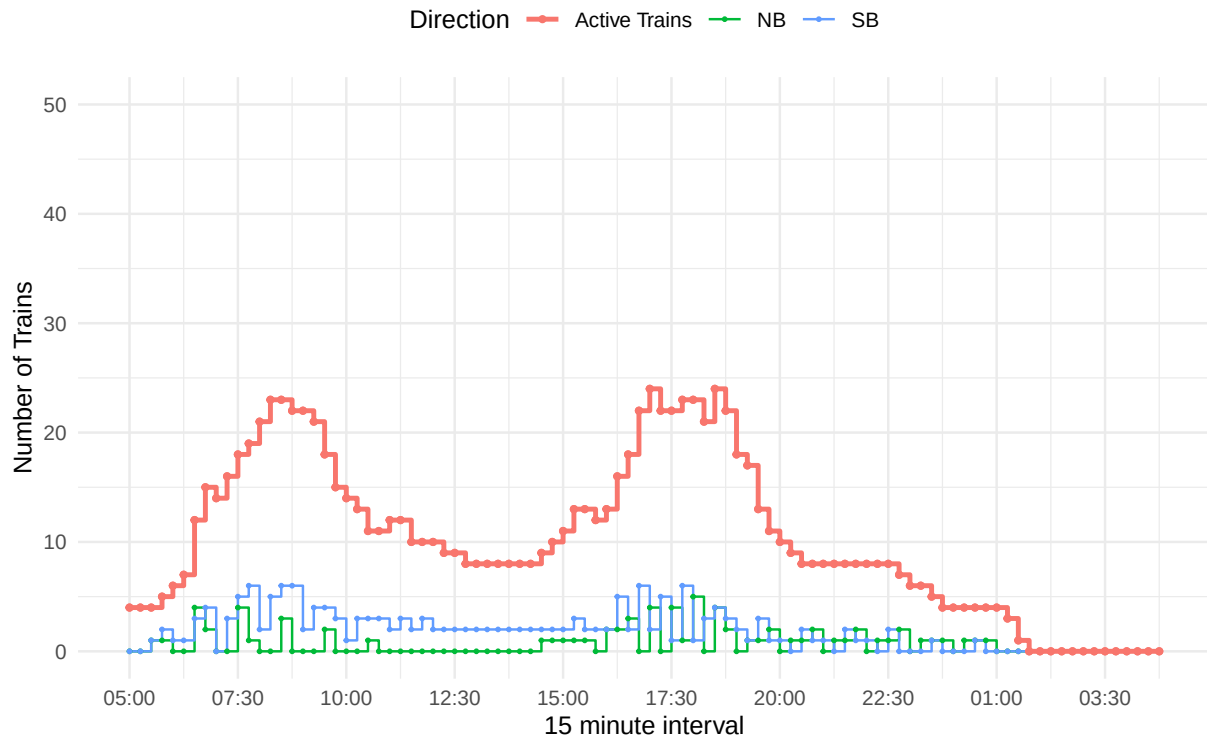
6.1 Effect of train capacity



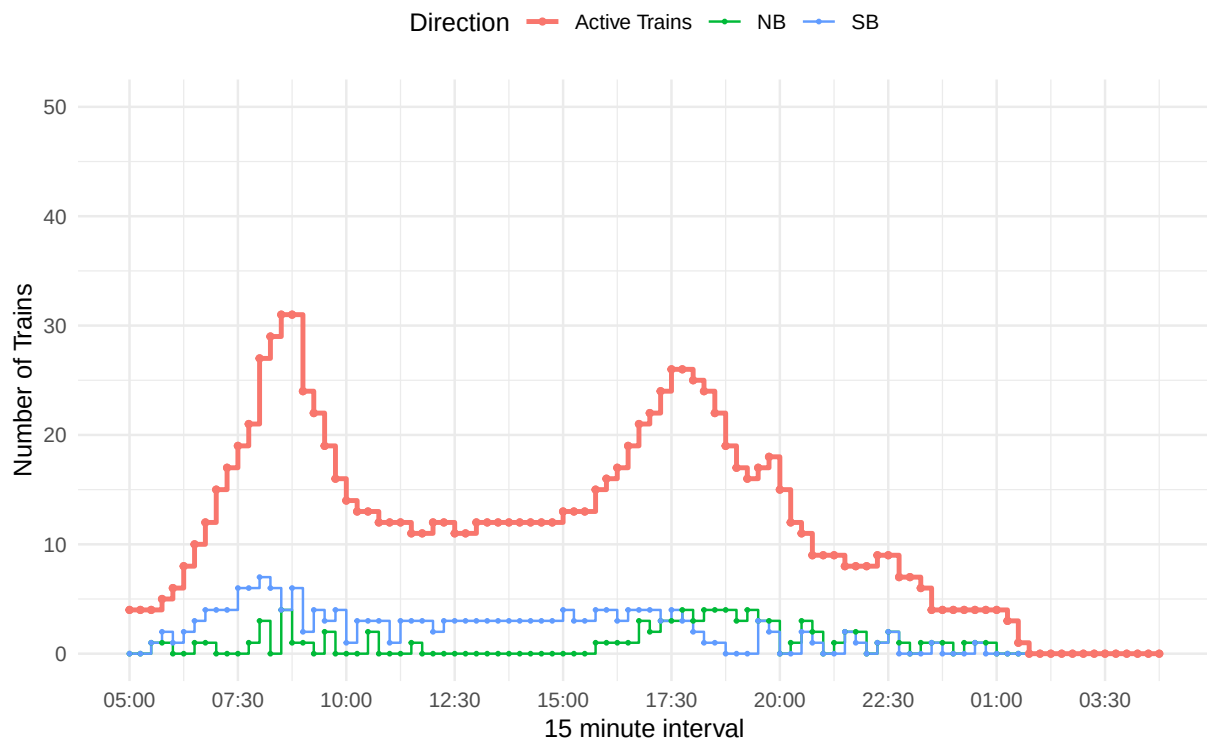
The plot suggests as capacity falls, the number of trains dispatched must rise in order to meet the constraint of zero overcrowding. At capacity 856, used in the model, the total number of trains dispatched was just below 250. But when capacity is reduced by 50% to 428 the total number of trains dispatched is just below 450 (an increase of around 200). Each time capacity is decreased by 12% the increase in the total number of dispatched trains rises slightly more. For example the increase in train dispatches when capacity decreases from 535 to 428 is much larger than the increase in trains when capacity is decreased from 856 to 749.

6.2 Effect on fleet utilisation

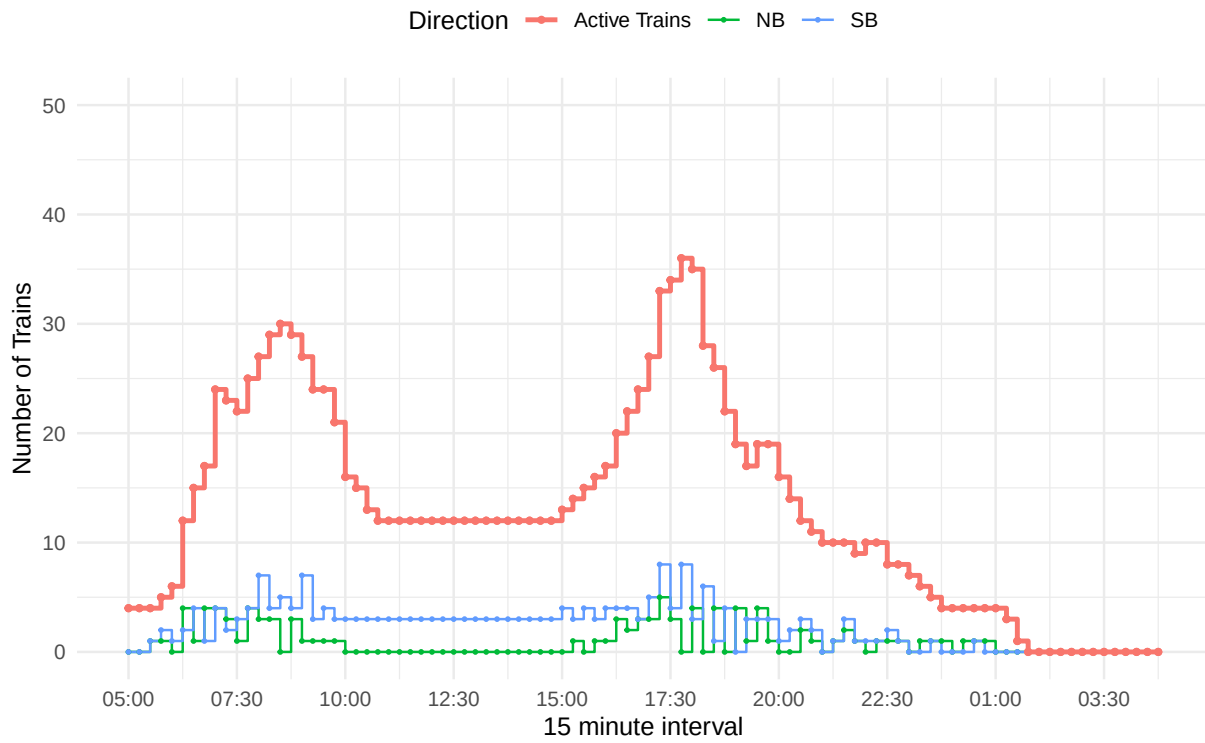
Optimised Train Dispatches at capacity 856



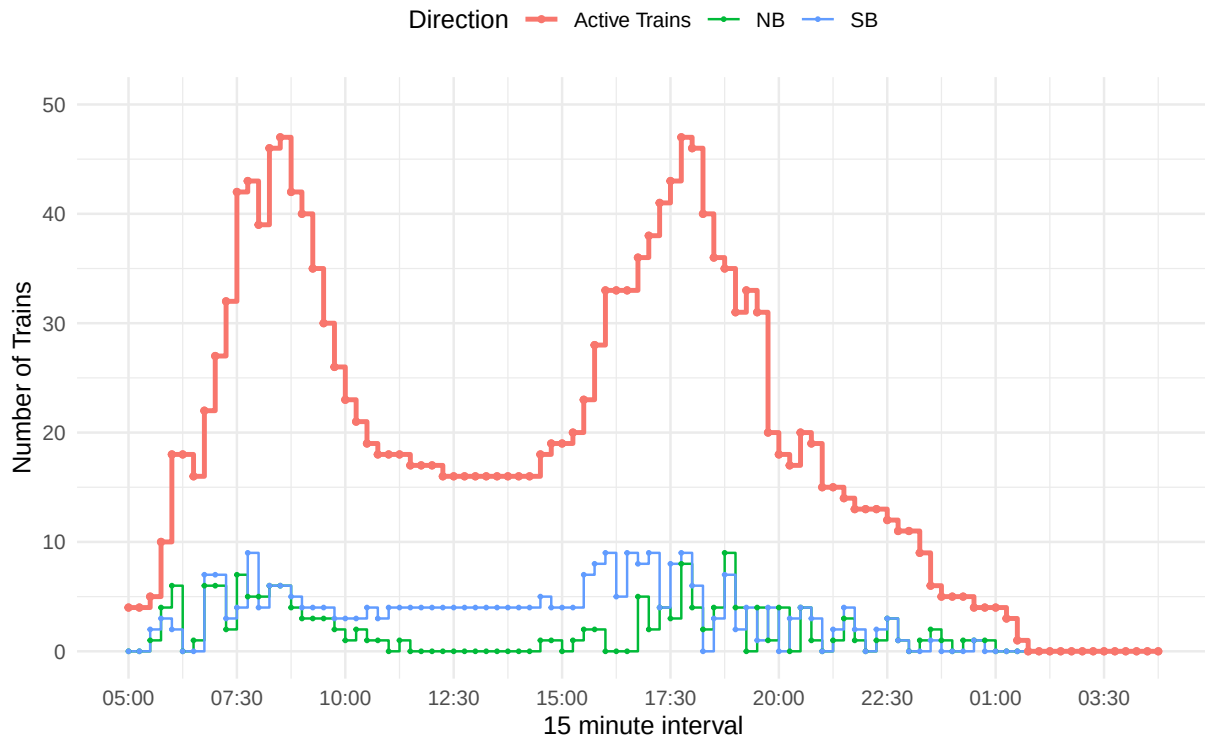
Optimised Train Dispatches at capacity 749



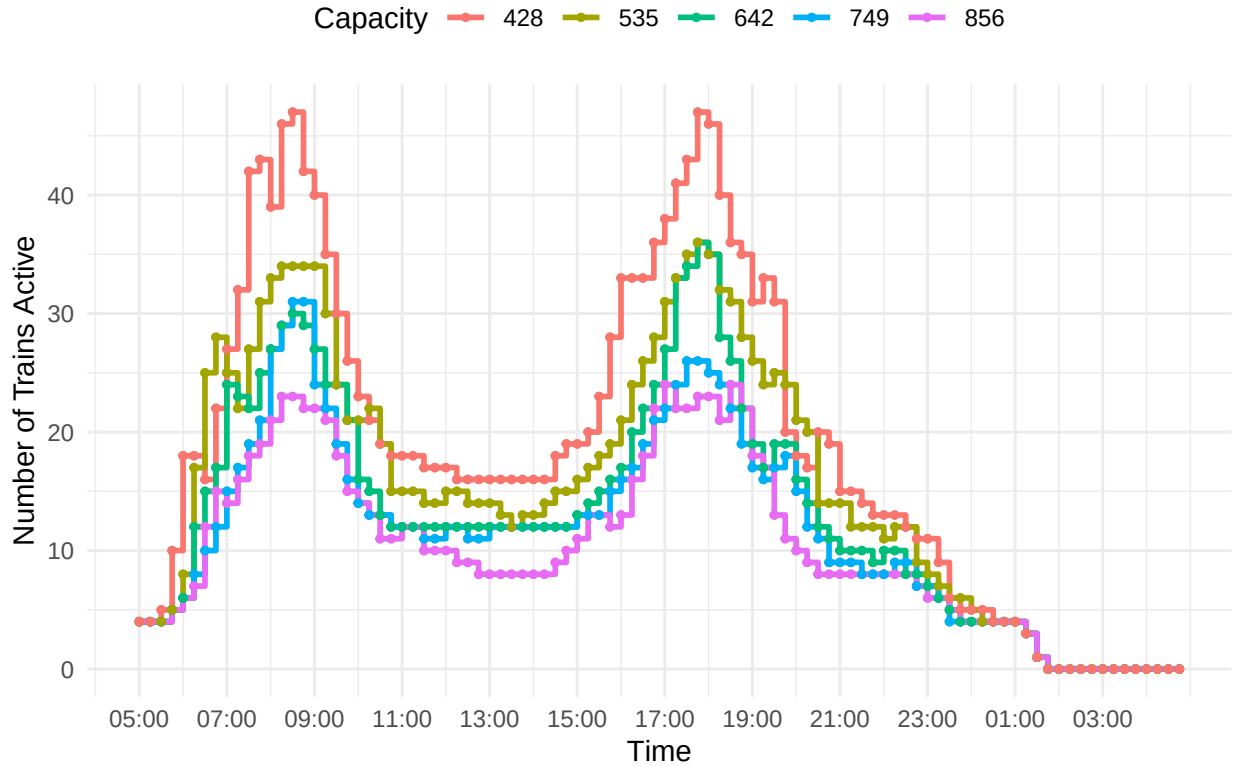
Optimised Train Dispatches at capacity 642



Optimised Train Dispatches at capacity 428



Minimised Active Trains for each Capacity and Interval



The first four plots show the number of active trains and trains dispatched in each direction across the capacities 856, 749, 642, 428. Meanwhile the plot below shows the number of active trains in each interval for the chosen capacities on one plot, providing a more direct comparison.

The plots show a steeper increase in active trains during peak times for the lower capacities. The individual plots also highlight how for the lower capacities a greater number of trains need to be dispatched in order to meet the increased passenger demand. For example at capacity 428 there are several intervals where 9 trains are dispatched in one direction (primarily SB) whilst the maximum number of trains dispatched in one direction during one interval for capacity 856 is 6. Even between peak times, the number of active trains is greater at lower capacities (at capacity 428 it falls to 16 but at capacity 856 it falls to 8).

However across the different capacities, the same pattern of the number of active trains peaking during peak times in the morning and evening can be observed, since they are all responding to the same link load data. The key difference is that at lower capacities a greater number of trains must be dispatched in order to meet this demand.

7. Limitations

8. Conclusion